

Using LibreOffice Calc

for Basic Statistical Analyses

Statistics is the discipline that involves the collection, organisation, display, analysis, interpretation and presentation of data. As data is today's oil, its statistical analysis is very crucial. With LibreOffice Calc, one can perform complex statistical or engineering analyses to give different results based on the data provided.

The following operations can be performed with the help of LibreOffice Calc:

- Sampling
- Descriptive statistics
- Analysis of variance (ANOVA)
- Correlation
- Covariance
- Exponential smoothing
- Moving average
- Regression
- Paired t-test
- F-test
- Z-test
- Chi-square test

A brief description of each of these operations is given in this article.

Sampling

Sampling is the selection of a subset (a statistical sample) of individuals from within a statistical population to estimate the characteristics of the whole population.

Sampling is done row-wise, i.e., the sampled data picks the whole line of the source table and puts the result into a line of the target table.

Sampling methods

Random: Picks exact sample size lines of the source table randomly.

Sample size: This is the number of lines sampled from the source table.

Periodic: This method picks lines at a pace defined by Period.

Period: This is the number of lines to skip periodically when sampling.

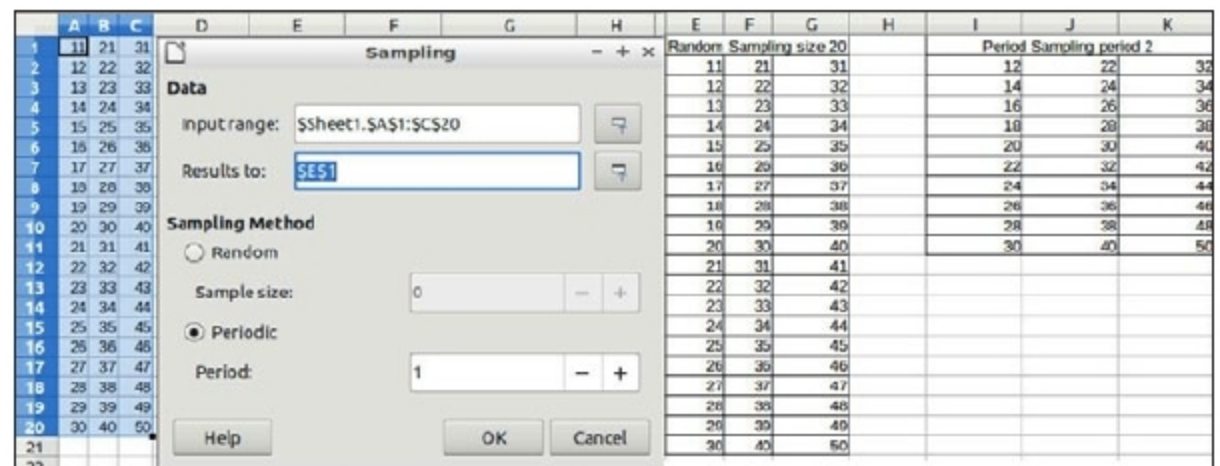


Figure 1: Sample source data, and sampling options and results

	Column 1	Column 2	Column 3
Mean	20.5	30.5	40.5
Standard Error	1.322075656	1.322075656	1.322075656
Mode	#VALUE!	#VALUE!	#VALUE!
Median	20.5	30.5	40.5
First Quartile	15.75	25.75	35.75
Third Quartile	25.25	35.25	45.25
Variance	35	35	35
Standard Deviation	5.916079783	5.916079783	5.916079783
Kurtosis	-1.2	-1.2	-1.2
Skewness	0	0	0
Range	19	19	19
Minimum	11	21	31
Maximum	30	40	50
Sum	410	610	810
Count	20	20	20

Figure 2: Sample data, options of descriptive statistics and results

To perform sampling, choose **Data** -> **Statistics** -> **Sampling**.

Sampling allows you to pick data from a source table/row and to fill a target table/row. The sampling can be random or periodical.

Data

Input: This refers to the range of the data to analyse.

Result: This refers to the top left cell of the range where the results are displayed.

In the input range, you have to specify the source of the data by selecting the columns. In 'Results' too, you have to specify the column name and row ID (e.g., \$E\$1 means column

E and Row 1). This convention is followed for other methods also.

Next, you have to specify the sampling method. Once done, click 'OK' and you can see the desired output (Figure 1).

Descriptive statistics

Descriptive statistics is a summary that quantitatively describes or summarises

